# LESSON PLAN

## LSF, STABILITY, A/C PERFORMANCE (6)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 25 |
| **Total No. of Lesson Plans** | 33 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Stability and control: static, dynamic stability. |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Stability and control: static, dynamic stability. definition, notation used Stability requirements, a/c controls and Maneuver |
| **5. Specific Objectives** | 6.1.4.1 Trainee will be able to describe aircraft stability in all the three axis. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of static stability definitions and dynamic stability concepts using IB and L & Ds; Instructor supervision required for concept verification. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What are the three axes of aircraft motion?
(b) Define the term 'stability' in the context of flight.
(c) List the primary flight controls and their associated axes.

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition of Static Stability<br>• Notation used for Stability<br>• Static Stability in Longitudinal Axis<br>• Static Stability in Lateral Axis<br>• Static Stability in Directional Axis | L & Ds | PPT, IB | 15 min |
| (b) | • Definition of Dynamic Stability<br>• Oscillatory Motion Characteristics<br>• Stability Requirements for Maneuver<br>• Role of A/C Controls in Stability<br>• Damping Effects in Dynamic Response | L & Ds | PPT, PDF | 20 min |
| (c) | • Comparison of Static and Dynamic Stability<br>• Practical Implications for Aircrew<br>• Summary of Stability Definitions<br>• Review of Notation Standards<br>• Link to Maneuver Requirements | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between static and dynamic stability.
(b) Describe how the aircraft responds to a disturbance in the longitudinal axis.
(c) Identify the notation used to represent stability derivatives.

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| **12. Home Assignment and Follow Up** | Prepare a brief note on how the concepts of static and dynamic stability learned in this lesson apply to the forces and moments discussed in LP 02. |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve., Longitudinal dynamic stability, | Refer to the master syllabus reference list | LP 26 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min